1.

a) (3 pts) From ECE210 we know that the time constant of exponential decay of the capacitor in an RC circuit is $\tau = RC$. Note that R remains the same as $R \equiv \frac{1}{G} = \frac{d}{A\sigma}$. From the lecture notes (chapter 10) we know that the capacitance of a parallel plate capacitor filled with a dielectric is

$$C = \epsilon \frac{A}{d},$$

where the conducting plates have surface areas A separated by a distance d. The capacitor with air filling has capacitance $C_1 = \epsilon_o \frac{A}{d}$, while the capacitance considering the slab is

$$C_2 = \epsilon_r \epsilon_o \frac{A}{d} = 1.25 \epsilon_o \frac{A}{d}.$$

Therefore the permittivity of the slab has to be $\epsilon = \epsilon_r \epsilon_o = 1.25\epsilon_o$ in order to increase the time constant in 25%.

b) (4 pts) We first calculate the total capacitance between the conductor plates. The electric field is $E_1 = D/\epsilon_o$ in the part filled with air, and $E_2 = D/\epsilon$ in the part filled with the dielectric. Then the total voltage drop between the plates is

$$V = E_1 \frac{d}{2} + E_2 \frac{d}{2} = \frac{d}{2} D \left(\frac{1}{\epsilon_o} + \frac{1}{\epsilon} \right).$$

From Gauss's law we obtain the displacement field $D = \frac{Q}{A}$, where Q is the total charge of the positive plate, and A is the area of the plate. This yields

$$C = \frac{Q}{V} = \frac{Q}{\frac{d}{2}\frac{Q}{A}\left(\frac{1}{\epsilon_o} + \frac{1}{\epsilon}\right)} = \frac{\epsilon_o \epsilon}{(\epsilon_o + \epsilon)} \frac{A}{d/2}.$$

This result can be obtained by calculating the capacitance of two capacitors in series with values $C_1 = \epsilon_o \frac{A}{d/2}$ and $C_2 = \epsilon \frac{A}{d/2}$, using $C = \frac{C_1 C_2}{C_1 + C_2}$. This capacitance is required to be 1.25 times the original one

$$\frac{\epsilon_o \epsilon}{(\epsilon_o + \epsilon)} \frac{A}{d/2} = 1.25 \epsilon_o \frac{A}{d},$$

which yields

$$\epsilon = \frac{5}{3} \epsilon_o.$$

Therefore the permittivity of the half-width slab has to be $\epsilon = \frac{5}{3}\epsilon_o$ in order to increase the capacitance by 25%.

c) (3 pts) From ECE210 we know that the time constant of exponential decay of an inductor in an RL circuit is $\tau = \frac{L}{R}$. From the ECE329 lecture notes (chapter 15) we know that the inductance of a solenoid filled with a material with permeability μ is

$$L = N^2 \mu A l,$$

where N is the number of loops per unit length, A is the cross-sectional area, and l is the length of the solenoid. Using this result the inductance of the solenoid with the rod is obtained as

$$L_2 = N^2 \mu_r \mu_o A l = 1.0005 N^2 \mu_o A l,$$

which yields

$$\mu = \mu_r \mu_o = 1.0005\mu_o = (1 + 0.0005)\mu_o.$$

Since $\mu_r = (1 + \chi_m)\mu_o = 1.0005\mu_o$, we notice that the magnetic susceptibility is $\chi_m = 0.0005$. Therefore with $\mu > \mu_o$, we can characterize the material as paramagnetic.

2. (5 pts) Referring to the hint, we should first find the divergence of the current density. Thus, we write

$$\nabla \cdot \mathbf{J} = \frac{\partial}{\partial x}\left(3z^2\right) + \frac{\partial}{\partial y}\left(2x^3 y\right) + \frac{\partial}{\partial z}\left(z(y - y_o)^2\right) = 2x^3 + (y - y_o)^2.$$

Then, taking the integral of both sides of the continuity equation $\frac{\partial \rho}{\partial t} = -\nabla \cdot \mathbf{J}$, we get

$$\rho(\mathbf{r}, t) = -\left(2x^3 + (y - y_o)^2\right) t + \rho_o \frac{\mathrm{C}}{\mathrm{m}^3}.$$

Evaluating $\rho(\mathbf{r}, t)$ at $\mathbf{r} = \mathbf{0}$ and given that $\rho_o = 0$ and $y_o = 1$, we find

$$\rho(\mathbf{0}, t) = -t \frac{\mathrm{C}}{\mathrm{m}^3}.$$

5.

a) (2 pts) The current in the outer loop can be calculated by $I_a = \frac{V}{R}$, where $R \equiv \frac{1}{G} = \frac{d}{A\sigma}$ as you may remember from Lecture 10. Thus, we write

$$I_a = \frac{V}{R} = \frac{V}{\frac{2\pi a}{\sigma \pi r^2}} = \frac{5}{\frac{2\times 1}{4\times 10^7 \times 0.001^2}} = 100\,\mathrm{A}.$$

b) (2 pts) Refering to the Page 8 in Lecture 13, the magnetic flux density generated by the outer loop only has z component along the z axis

$$B_z = \frac{\mu_o I_a a^2}{2(a^2 + z^2)^{\frac{3}{2}}}.$$

At the origin, $B_z(z = 0) = \frac{\mu_o I_a}{2a} \frac{Wb}{\mathrm{m}^2}$. Since $b \ll a$, we assume that the B_z across the inner loop is constant. Thus,

$$\Psi_{a\to b} = \int_S \mathbf{B} \cdot d\mathbf{S} = \int_S B_z(z = 0)\hat{z} \cdot \hat{z} dS = \frac{\mu_o I_a}{2a}\pi b^2 = \frac{\mu_o \pi b^2}{2a} I_a\,\mathrm{Wb}.$$

c) (2 pts) The numerical value of $L_{a\to b}$ is

$$L_{a\to b} = \frac{\Psi_{a\to b}}{I_a} = \frac{\mu_o \pi b^2}{2a}\,\mathrm{H}.$$

d) (4 pts) Applying induced emf formula $\mathcal{E} = -\frac{d\Psi}{dt}$, we get

$$\mathcal{E}_{a\to b} = -\frac{d\Psi_{a\to b}}{dt} = -\frac{d}{dt}\int_S B \cdot dS = -\frac{d}{dt}(B_z A_b) = -A_b \frac{d}{dt} B_z.$$

Here, B_z is given by

$$B_z = \frac{\mu_o I_a a^2}{2(a^2 + z^2)^{\frac{3}{2}}},$$

Therefore, we can write

$$\frac{d}{dt}B_z = -\frac{3}{2}\frac{\mu_o I_a a^2 z}{(a^2+z^2)^{\frac{5}{2}}}\frac{d}{dt}z = -\frac{3}{2}\frac{\mu_o I_a a^2 z}{(a^2+z^2)^{\frac{5}{2}}}v_z,$$

from which we obtain

$$\mathcal{E}_{a\to b} = \frac{3}{2}\frac{\mu_o I_a a^2 z\pi b^2}{(a^2+z^2)^{\frac{5}{2}}}\,(\mathrm{V}).$$

If we consider, $z = v_z t$, we can write $\mathcal{E}_{a\to b}$ as a function of time t as (here $v_z = 1$)

$$\mathcal{E}_{a\to b}(t) = \frac{3}{2}\frac{\mu_o I_a a^2 t\pi b^2}{(a^2+t^2)^{\frac{5}{2}}}\,(\mathrm{V}).$$

e) (5pts) As the induced emf $\mathcal{E}$ is given by $\mathcal{E} = -\frac{d\Psi_t}{dt}$, also the current I by $I = \frac{\mathcal{E}}{R}$. Then, the induced current in the inner loop will be

$$I_b = -\frac{1}{R}\frac{d\Psi_t}{dt},$$

where $\Psi_t = \Psi_{a\to b} + \Psi_s$, $\Psi_{a\to b} = -\int \mathcal{E}_{a\to b}(t)\,dt$ and $\Psi_s = LI_b$ ($L = 0.25\,\mu\mathrm{H}$). Thus, we can write

$$I_b = -\frac{1}{R}\frac{d\Psi_{a\to b}}{dt} - \frac{L}{R}\frac{dI_b}{dt}.$$

Rearranging the above equation, it will give us the differential equation as

$$\frac{dI_b}{dt} + \frac{R}{L}I_b = -\frac{1}{L}\frac{d\Psi_{a\to b}}{dt} = \frac{\mathcal{E}_{a\to b}(t)}{L},$$

where right hand side is the emf $\mathcal{E}_{a\to b}(t)$ of part (d) divided by L.